Application Help

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SAP Business Network Freight Collaboration | LBN

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SAP Business Network Freight Collaboration

SAP Business Network Freight Collaboration improves supply chain efficiency by connecting business partners on a collaborative network that supports jointly managing transactions, exchanging documents, and sharing insights across the value chain. For instance, upon invitation from a shipper, a carrier can onboard onto the network and collaborate with shippers on freight tendering, invoicing and milestone reporting.

SAP Business Network Freight Collaboration runs on SAP Business Technology Platform (SAP BTP), utilizing one of the following infrastructures:

- Amazon Web Services (AWS) infrastructure, hosted in a data center in Frankfurt, Germany
- Microsoft Azure infrastructure, hosted in a data center in Washington state, US
- Google Cloud infrastructure, hosted in a data center in Dammam, Saudi Arabia

The data used in the solution is also stored in the respective data center.

Information for shippers:

Collaborative managed transactions supported in SAP Business Network Freight Collaboration are tailored to the relevant SAP backend systems. For freight collaboration scenarios, SAP Business Network Freight Collaboration offers out-of-the-box connectivity to shippers with the following systems:

- SAP Transportation Management (on-premise) or Transportation Management embedded in SAP S/4HANA (on-premise)
- SAP S/4HANA Cloud Private Edition
- SAP Logistics Management

ℹ Note

The following documentation uses the term *SAP S/4HANA* to refer to both SAP S/4HANA Cloud Private Edition and SAP S/4HANA (on-premise). If any of the information provided applies to only SAP S/4HANA Cloud Private Edition or SAP S/4HANA, this will be clearly specified.

For more information about supported releases, see SAP Note 2620322. In addition, out-of-the box connectivity is delivered for truck carrier networks, including project44 for visibility services and B2B carrier connectivity for freight tendering and subcontracting. Selected digital freight forwarders support on-demand road transportation with real-time pricing and subcontracting.

To learn more about the SAP Business Network add-ons for project44 visibility services, see the Feature Scope Description for project44 Add-Ons and the Configuration and Application Guide for project44 Add-Ons.

Shipper Apps and APIs

With SAP Business Network Freight Collaboration, you as a shipper can use the apps to get started on the network, including connecting your systems (for example a transportation management system) to the network. Additionally, you can perform your daily tasks in the areas of dock appointment management, freight order tracking, freight settlement, and dispute handling.

ℹ Note

These apps are also suitable for logistics service providers that are performing the role of a shipper.

Prerequisites

To use a new app, you must add one or several roles to your role collection. The required role templates and application identifiers are described in the Available Roles for Shippers section.

Shipper APIs

As a shipper, you can use the APIs in SAP Business Network Freight Collaboration to perform your daily tasks in the areas of freight contracting, freight booking, freight tracking, dock appointment scheduling, and freight settlement.

For more information, see https://help.sap.com/doc/e1b2026259d84fc29e6e6760da815011/LBN/en-US/7ccd8fd6e75249b991c8d62706fe90ff.html.

General Settings

The **General Settings** app enables you as the shipper admin to configure the general settings related to freight collaboration at an ordering party level.

You can also set up tolerance rules that would enable the settlement clerk at the shipper organization to auto-approve certain amounts by applying the tolerance rules to disputes created from invoices or self-billing documents.

When an invoice contains discrepancies that would result in a dispute, you can define and assign tolerance ranges and tolerance groups so that you do not have to check each order manually. The system can then approve charge lines automatically. Lines that are not approved must be reviewed manually by the user.

ℹ Note

To use this app, you must add one or several roles to your role collection. The required role templates and application identifiers are described in the Available Roles for Shippers section of the Administration Guide for Shippers.

This app contains the following main areas:

- **Define Appointment and Dispute Settings**

  You can configure the appointment and dispute settings for an ordering party in this tab. For more information, see Appointment and Dispute Settings.

- **Define Tolerance Range**

  You can define the tolerance range itself by creating an upper limit and a lower limit. The tolerance values can be specified in either amounts (e.g. lower limit of EUR 10, upper limit of EUR 15) or percentages (e.g. lower limit of 2%, upper limit of 2%).

  ### ℹ Note

  You must specify a lower limit and an upper limit, either of which can be expressed as an amount or a percentage.

- **Define Tolerance Groups**

  Once you have defined the tolerance range, you can group them together. You can define item level tolerance ranges for each charge code. You can add a tolerance range for the rate amount and final amount for each charge code.

  You can also define the tolerance ranges for the total document amount proposed by the carrier. In this scenario, for each currency, you can specify the tolerance range.

  For more information on how tolerance is applied on dispute documents, see Applying Tolerance Check.

Appointment and Dispute Settings

SAP Business Network for Logistics enables you to configure the dock appointment and dispute settings for an ordering party. As part of the dock appointment settings, you can also define the guest user access settings.

Guest User Access

In the **General Settings** app, in the **Define Appointment and Dispute Settings** tab, you can choose the **Guest Access** button to define the type of URL you want to send to the guest carrier user. You can choose one of the following options:

- Dynamic URL

  When you select this option, the system sends a URL to the carrier which is specific to the document you send to the carrier for dock appointment scheduling. The carrier can enter the document reference to view the document.

- Static URL

  When you select this option, the system sends a URL to the carrier which is generic. The carrier can enter the following details and view the document:

  - Document Type and ID
  - Email ID
  - User Name
  - Company Name

  We recommend you to select the dynamic URL by default. If you select the static URL, anyone with the link and valid document reference can access the document.

Note

The system applies the settings you specify here to all the ordering parties. Once you specify the above settings, the system displays the type of URL you have chosen on the header area in the tab view. In the detailed view, you cannot view this information.

If you select the dynamic URL option and later change the setting to static URL, the guest user can use both the dynamic and static URLs to access the document. However, if you select the static URL and later change the setting to dynamic URL, the guest user can only use the latest dynamic URL to access the document.

Ordering Party Settings

To define the ordering party settings, in the **General Settings** app, in the **Define Appointment and Dispute Settings** tab, choose the **Add Ordering Party** button.

Note

The system only displays those ordering parties that you have maintained in the **Manage Locations** app with the object type as **Business Partner**.

Once you add an ordering party, you can define the following settings:

Dock Appointment Scheduling

You can enable one of the following settings for the dock appointment creation process in the SAP Business Network for Logistics system:

- **On Freight Order Creation**

The system creates an appointment when a freight order is created. In other words, the system creates an appointment for a freight order that is yet to be confirmed.

- **On Freight Order Confirmation**

  The system creates an appointment only when a freight order is confirmed.

- **On Appointment API**

  The system creates an appointment based on the appointment API and not on the freight order.

> **Note**
> To add an ordering party, you must have already created a location in the **Manage Locations** app with the **Object Type** as **Business Partner**.

If you select **On Freight Order Creation** or **On Freight Order Confirmation**, the system still allows you to create appointments using the Appointment Provider API (`GenericAppointment_In`). If you select **On Appointment API**, appointments can only be created using the API.

Settlement

Dispute Management in External System

You can enable or disable dispute management in external system for freight orders and freight bookings.

Enable for Freight Order

When you enable this setting for freight orders, then the disputes created for the subsequent documents will be created in your external system.

Enable for Freight Booking

When you enable this setting for freight bookings, then the disputes created for the subsequent documents will be created in your external system.

> **Note**
> It can take up to 15 minutes until any change is reflected in the carrier's **Manage Invoices** app.

Enable Automatic Submission of Disputes

You can enable or disable automatic approval of disputes. If you have enabled this setting, and if the dispute is within the specified tolerance range, the system automatically approves the disputes sent by your carrier. If the dispute is not within the specified tolerance range, you must manually approve the dispute.

Note that if you have defined both document tolerance and item tolerance, then both tolerances must be within the specified tolerance range for the dispute to be automatically submitted. If you have defined only document tolerance or item tolerance, then the respective tolerance must be within the specified tolerance range for the dispute to be automatically submitted.

Enable Mandatory Attachments for Invoice Submission

You can enable or disable mandatory attachments during carrier invoice submission. If you enable this setting, during carrier invoicing, the carrier must mandatorily add attachments before submitting the document. If you disable this setting, adding an attachment is optional for the carriers.

Enable Subsequent Invoice with Same Charge Type

You can enable or disable subsequent invoice with same charge type. When you enable this setting, the carrier can add a subsequent charge for a charge type which has already been invoiced. If you disable this setting, the carrier cannot add a subsequent charge.

Reference Documents

You can specify whether you want to send reference documents to external system as notes. If you enable this setting, you can also specify the note type for reference documents in the field that appears below. This note type that you specify here will be used to send the reference documents as a note.

> **Note**
>
> Use a note type that is different from the default note type as the system uses the default note type to send the carrier notes to your external system.

Enable Tax Code Validation

When you enable this setting, if the carrier enters a tax code in the invoice, the system validates this tax code with the tax codes that you have maintained in the **Assign Charge Codes** app.

Tolerance Group Assignment

You can assign one tolerance group to each individual carrier that you have onboarded on the freight collaboration option for SAP Business Network for Logistics. When you assign a tolerance group to a carrier, the tolerance group should be sufficient for consumption in terms of charge lines and the respective currencies. For example, if you have a very good relationship with a particular carrier, you can define a group in which all the charge codes are automatically approved.

Approval Conditions

You can enable a multi-step dispute approval process within your organization before submitting the dispute to the carrier.

A dispute document can either be approved in a single step by a basic user at the first level or sent to supervisors or managers at the next levels for review based on the settings that you specify. The additional approval steps are enabled only when the disputed amount exceeds the threshold amount or percentage specified the settings.

Based on the range of the disputed amount, you can define multiple approver groups for various dispute amount ranges to approve the dispute. You can specify the approval conditions for an ordering party as follows:

1. Choose **Add** to add a new approval condition.

   Add **Approver Conditions** pane appears on the right.

2. In the **Approver Group** section, specify the following:

   - **Scenario**

     Displays dispute scenario by default.

   - **Approver Group**

     Specify the name of the approver group. The name that you specify here must be the same as you have specified in the user group in IAS Portal.

     You can assign all the users for this approver group to the IAS user group. For more information on assigning user groups, see Organize Roles Together and Assign Them to Users.

     > **Note**
     >
     > This is a mandatory field.
     >
     > You can specify alpha numeric values.

Do not specify commas and spaces.

You can only specify _ (underscore) and – (hyphen) special characters.

- **Notification Language**

  Select a language, from the drop-down menu, in which you want to receive the email notification. An email notification is sent to the approvers in this group whenever a dispute requires their approval.

- **Notification Emails**

  Specify the email ID to which the email notification must be sent. You can specify multiple email IDs by choosing the add icon.

  Note that the system only sends email notifications about the dispute approval to these users. Not all these users may have the authorization to approve or reject the disputes. You must maintain those settings in the IAS portal.

3. In the **Conditions** section, specify the following:

   - **Currency**

     Select the same currency that you have specified for the document currency, from the drop-down menu.

   - **Threshold Range**

     Select a threshold amount or a threshold percentage from the drop-down menu. Specify a threshold amount or percentage as appropriate.

   - **Purchasing Group**

     Specify the name of the purchasing group.

     Note

     - For an ordering party, you must specify the same currency and threshold range factor (amount or percentage) across all approval conditions. In other words, when you create subsequent approval conditions, you must specify the same currency and threshold range factor (amount or percentage) that you specify in your first approval condition.
     - You cannot enter the same combination of condition attributes in two approval groups. In other words, you cannot specify the same currency, threshold range, and purchasing group combination that you specified in the previous entry.

Administrative Data

Displays the administrative data such as created on, created by, changed on, and changed by.

Applying Tolerance Check

When a dispute is created or updated in the carrier tenant for a freight order or invoice and then sent to you for review, a tolerance check is applied based on your tolerance settings. You can enable tolerance for the entire document amount or the individual charge item amount.

Tolerance Check Formula

The system uses the following formulae to check whether the amounts are within the specified tolerance range:

- **For Tolerance Amounts**

If you have maintained the tolerance range in amounts, the system uses the following formulae for calculation:

Freight Order Amount – Lower Tolerance Amount

Freight Order Amount + Upper Tolerance Amount

If the amount proposed by the carrier is in between the above values, then the tolerance check is approved.

If the freight order amount is USD 50 and the upper limit is USD 5 and lower limit is USD 5, then if the carrier specifies the amount to be within USD 45(50-5) and USD 55(50+5) then it passes tolerance check.

- **For Tolerance Percentages**

  If you have maintained the tolerance range in percentages, the system uses the following formulae for calculation:

  Freight Order Amount – (Lower Tolerance Percentage * Freight Order Amount)

  Freight Order Amount + (Upper Tolerance Percentage * Freight Order Amount)

  If the amount proposed by the carrier is in between the above values, then the tolerance check is approved.

  For example, if the freight order amount is USD 50 and the upper limit is 5 % and lower limit is 5 %, then if the carrier specifies the amount to be within USD 47.5(50-2.5) and USD 52.5(50+2.5) then it passes tolerance check.

Document Tolerances

If you have defined document tolerances in your tolerance group for a carrier, a tolerance check is applied on the total document amount proposed by the carrier. In other words, if the carrier sends a disputed document to you and the total freight document amount proposed by the carrier on the header area falls within the defined tolerance range, then the system updates the information button on the header dispute status with this information.

Item Tolerances

If you have defined item tolerances in your tolerance group for a carrier, the system applies a tolerance check according to the following criteria:

- If the charge codes listed in a freight document are present in the defined tolerance group, the tolerance check is applied for those charge codes.
- If the charge codes are not specified in that group, no tolerance check is applied and you must approve those lines manually.

For a charge code that matches the group, the currencies are then checked. In the **General Settings** app, the **Rate Amount** and the **Final Amount**, for example, can be maintained in different currencies. If the currencies match, a second check is applied according to a defined formula.

For a given currency, you can maintain a limit as either an amount or a percentage, and different formulas are applied accordingly. If the carrier's proposed value in the invoice lies within the start and end limits according to the respective formula, the charge line is considered to be within tolerance.

Every rate amount and final amount that falls within the tolerance limit is assumed to be approved by the system's tolerance check. The line is then marked with a green bar and receives the status **Approved by tolerance check**. This means you do not have to approve these lines manually.

If the tolerance check does not meet your expectations, you can override it by clicking *(Edit)*and entering your changes. If you do so, however, you must consider the following:

- If you edit the line, the tolerance check will not be applied to that line in subsequent rounds. (However, if you do not edit the line and it does not pass the tolerance check in the first round, the check can be applied in all subsequent rounds of the dispute life cycle process.)

- If all the lines are approved by the tolerance check, the dispute status remains as **For review**. (It only changes to **Resolved** if you click **Submit Dispute**.)

If a line is dependent on other lines and the other lines are approved by the tolerance check, the dependent line is not necessarily approved by the check unless it falls within the tolerance limits.

If you add a new line, the same tolerance check is applicable. In this case, however, the freight order value is considered as 0. Only if it matches the currencies and falls within the tolerance limits is the new line also considered as approved by the tolerance check.

Combination of Settings

If you have specified the settings for both document tolerance and item tolerance, then the system checks the tolerances at both levels in the disputed document and updates the status as follows:

- For the charge lines within tolerance range, the status is updated as **Approved by Tolerance Check**.
- For the charge lines outside tolerance range, there is no status update.
- For the charge lines for which tolerance is not defined, there is no status update.
- On the header status, in the information button, a message is displayed whether the tolerance is within the set tolerance range or not.

If you have specified the settings only for document tolerance or item tolerance, then the system checks the tolerances as appropriate based on your settings.

Example

You have specified the following settings in the tolerance group:

Document Tolerance Amount

Currency	Tolerance Definition	Selected Tolerance Range
EUR	Example1	10.00 EUR to 100.00 EUR

Item Tolerance Amount

Charge Code	Rate Amount Tolerance Range	Final Amount Tolerance Range
ADMIN_FEE	10.00 EUR to 50.00 EUR	10.00 EUR to 50.00 EUR
INSPECTION_FEE	10.00 EUR to 50.00 EUR	10.00 EUR to 50.00 EUR

Freight document amount is EUR 100.

If the carrier sends a disputed document to you, the system checks the tolerances at both levels in the document as you have maintained both document tolerance and item tolerance.

If the total document amount proposed by carrier is within EUR 90 (100-10) to EUR 200 (100+100), the document tolerance check is approved.

If both the rate amount and the final amount of the administration fee and inspection fee are within EUR 90 (100-10) to EUR 150 (100+50), then the system updates the status of these charge lines as **Approved by Tolerance Check** and the charge line is highlighted in green.

ℹ Note

- If the subsequent invoice contains a charge that is already invoiced, the subsequent invoice is not applicable for document tolerance.
- If the subsequent invoice contains a charge that is already invoiced, that charge is not applicable for item tolerance.

Manage Locations

ℹ Note

To use this app, you must add one or several roles to your role collection. The required role templates and application identifiers are described in the Available Roles for Shippers section of the Administration Guide for Shippers.

You can use the Manage Locations app to manage location master data either manually replicated from another system or created from scratch using the app. Location master data can then be consumed by other apps, such as Docking Location Settings, in SAP Business Network Freight Collaboration.

You can set up a location replication using the Data Replication Framework (DRF) in the SAP S4/HANA source system. For more information, see the *Business Configuration* section of the *Administration Guide for Shippers*.

Location master data consists of not only the address information, but also the business purpose of a location. Additionally, the master data of a location can include one or more alternative IDs so that the location becomes cross-platform identifiable.

The main features of this app include:

- View a location
- Create a location
- Upload locations
- Edit a location

ℹ Note

- The Manage Locations app implements all security features and data protection-relevant functions documented in the "Security and Data Protection and Privacy" section of the *Administration Guide*.
- A location master data record enters the blocking period if it has not been accessed for three consecutive years. The blocking period lasts for another three years, during which only audit specialists can access the record. The record will then be automatically deleted when the end of the blocking period is reached.

Creating a Location

To create a location, proceed as follows:

1. Click **Create** on the upper-right corner of the **Locations** table. The **Create New Location** page opens.
2. On the **Create New Location** page, enter the following details:
    - **Source**: Source of the location. Valid values are:
        - **SAP SCM Logistic Location**: Locations from one of the following systems: